

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		Any 2 × (1) from: Heater 1 uses more energy / heater 1 is less efficient / heater 2 uses less energy / heater 2 is more efficient (1) Heater 1 <u>heats</u> up more water / heater 2 <u>heats</u> up less water / heater 1 <u>heats</u> up 150 and heater 2 <u>heats</u> up 50 litres / heater 2 only <u>heats</u> up 50 litres / 50 litres is closer to 40 litres (1) Heater 1 wastes more hot water / heater 2 wastes less hot water / heater 1 wastes 110 litres of water and heater 2 wastes 10 litres of hot water / heater 2 only wastes 10 litres of hot water (1) Heater 1 costs more money / Heater 2 costs less money / Heater 1 costs 132p and heater 2 costs 44p / heater 2 only costs 44p (1) Heater 1 takes more time / heater 2 takes less time / heater 1 takes 3 hours to heat the water and heater 2 takes half an hour / heater 2 only takes half an hour to heat the water (1) (So disagree) references to payback time neutral			2	2		
	(b)			Furred up / blocked pipes in boiler /rust /decreased water flow/ less efficient / takes longer to heat up the water / harder to heat up the water / hard water has a higher boiling point (1) (due to build up of) limescale / calcium or magnesium carbonates / scum / calcium and magnesium ions (1) Neutral: causes damage / makes bill go up	2			2		